

# Prompt-Based Jailbreaking of Leading LLM Chatbots: A Survey of Attacks and Defenses

Brynn Knowlton , Jovani Campa , David Solis Gallo , Khalil Dajani ,  
and Nabeel Alzahrani 

**Abstract**—Generative AI systems—particularly large language models (LLMs)—remain vulnerable to jailbreak attacks: adversarial prompts that bypass safeguards and elicit unsafe or restricted outputs. This survey synthesizes jailbreak research from 2023–2025, covering attack methods, defense strategies, and evaluation frameworks. Jailbreak techniques are grouped into five main categories: prompt-based injections, role-play conditioning, multi-turn dialogue, multilingual or multimodal exploits, and optimization-driven pipelines. We also examine discovery methods such as human red teaming and automated attacker systems (e.g., AutoJailbreak, LLMStinger), and review defense mechanisms including supervised fine-tuning (SFT), reinforcement learning from human feedback (RLHF), adversarial fine-tuning, and output or pipeline filtering. Standardized benchmarks—such as PromptBench, JailbreakBench, and multimodal evaluation suites—are analyzed in terms of reproducibility, coverage, and robustness testing. Persistent issues of generalization, reproducibility, and scalability are discussed, along with trade-offs between safety alignment and model utility. This paper provides an up-to-date reference clarifying the evolving landscape of jailbreak attacks and defenses in LLMs.

**Impact Statement**—Large language models are now used widely in education, healthcare, and security, yet they remain prone to jailbreak attacks that can defeat alignment controls. This paper reviews over one hundred recent studies, organizing jailbreak methods, defenses, and benchmarks into a coherent overview. The survey is relevant to AI safety and governance communities, summarizing how standardized evaluation and layered defense pipelines support reproducible robustness assessment. By consolidating recent findings and highlighting recurring challenges in generalization, reproducibility, and scalability, this work offers a concise reference for building safer and more transparent generative AI systems.

**Index Terms**—Adversarial machine learning, Explainable AI, Foundation models, Generative AI, Large Language Models, Prompt engineering

## I. INTRODUCTION

Generative AI systems powered by large language models (LLMs)—such as ChatGPT [1], Claude [2], and Gemini [3]—have reshaped areas including education, healthcare, and cybersecurity. Yet their rapid adoption has exposed a consistent safety issue: jailbreak attacks, where adversarial prompts override safeguards and produce unsafe or restricted outputs [4], [5]. Approaches range from “Do Anything Now” (DAN) role-play prompts [4] and adversarial suffix manipulations [6] to multilingual exploits exploiting gaps in safety coverage [7]. Automated pipelines like AutoJailbreak [5] and LLMStinger [8] show that adversarial prompt generation can

now be scaled and optimized, exposing vulnerabilities even in production-grade models. This survey reviews jailbreak attacks and defenses for LLMs, focusing on work from 2023–2025—a period marked by standardized benchmarks such as PromptBench [9] and JailbreakBench [10]. The goal is not to propose new defenses but to consolidate and clarify existing ones. We outline five recurring attack types—prompt-based, role-play, multi-turn, multilingual or multimodal, and optimization-driven—and summarize key defense methods (SFT, RLHF, adversarial fine-tuning, and output or pipeline filtering). Supervised fine-tuning (SFT) aligns models using curated instruction–response pairs, while reinforcement learning from human feedback (RLHF) uses preference-based reward models to refine model behavior. We also discuss benchmark evaluation frameworks and common research challenges: generalization, reproducibility, and scalability. The remainder of the paper is structured as follows: Section II describes the methodology; Section III details the taxonomy of jailbreak attacks; Section IV covers discovery paradigms; Sections V–VII review evaluation and defenses; Section VIII discusses benchmarks; Section IX outlines open problems; Section X describes threats to validity; and Section XI concludes. Beyond consolidating prior work, this survey reveals a unifying pattern across recent jailbreak research: jailbreak attacks increasingly function as optimization processes—whether through evolutionary refinement, reinforcement learning, or LLM-driven adversarial generation—while most existing defenses remain static, rule-based, or narrowly fine-tuned. This widening “optimization–alignment gap” helps explain why new jailbreaks routinely succeed even against models with advanced alignment methods such as RLHF or Constitutional AI. By synthesizing evidence across 110 studies, our survey provides a perspective that is not readily apparent from individual papers: jailbreak vulnerability arises not from isolated technical failures, but from systemic asymmetry between rapidly evolving attack search spaces and comparatively rigid defense mechanisms.

## II. METHODOLOGY

### A. Source Selection

This survey synthesizes over 500 peer-reviewed papers and technical preprints published between 2023 and 2025 that examine jailbreak attacks, defenses, or evaluation frameworks for generative AI. Relevant works were identified through systematic keyword searches (“LLM jailbreak,” “prompt injection,” “adversarial suffix,” “automated jailbreak discovery”) across IEEE Xplore [11], arXiv [12], and the ACM Digital Library [13]. These repositories capture most reproducible,

The authors are with the School of Computer Science and Engineering at California State University San Bernardino (CSUSB) San Bernardino, California, USA

state-of-the-art research on AI security and language modeling. The focus on 2023–2025 reflects the period of fastest advancement in LLM jailbreak research, during which reproducible attacker pipelines, standard benchmarks, and large-scale evaluations emerged. Because LLM safety architectures evolve quickly, maintaining a focused temporal scope ensures that the synthesis represents the methods and vulnerabilities relevant to current-generation systems. Earlier adversarial-learning research, while foundational, predates current versions of Large Language Models and so is less relevant to current research.

### B. Inclusion Criteria and Refinement

The initial search returned roughly 500 records. Screening occurred in two stages. First, duplicates, non-empirical works, and papers unrelated to generative architectures were removed. Second, remaining papers were reviewed for relevance to jailbreak attacks, discovery methods, defenses, or evaluation frameworks. Studies were included if they described methods and data transparently and provided quantitative or benchmarked evaluation on LLMs. When multiple versions existed, the latest or most complete was used. Balance across model families, themes, and approaches was prioritized through qualitative review rather than numeric weighting. To ensure representativeness, the refined set was reviewed to verify that all five analytical dimensions—attack techniques, discovery strategies, defense mechanisms, evaluation benchmarks, and mitigation approaches—were covered without dominance by any single area. No fixed quotas were applied; balance was achieved through iterative assessment of coverage across models and research threads. The result is a curated set of 110 representative studies forming the empirical foundation of this survey. The overall methodological workflow for literature collection and refinement is illustrated in Fig. 1.

### C. Representativeness and Reproducibility

The final 110 works capture a balanced cross-section of jailbreak research across major model types and methodologies, including both academic and applied studies. Representativeness was monitored qualitatively to ensure proportionate coverage of prompt-injection, suffix-based defenses, red-teaming, and benchmark development. Since all works appear in the reference list, the citations themselves provide a transparent record of sources. Because all representative works appear in the reference list, the citations themselves serve as a transparent record of the analyzed literature. The 110 representative works forming the basis of this survey correspond directly to this reference list, providing a transparent and reproducible record of all included studies.

### D. Background on Alignment and Safety

Large language models are typically trained to balance two broad objectives: being helpful to users by providing informative responses, and being safe by avoiding harmful or restricted content. Alignment techniques—such as providing curated examples of desirable behavior, applying preference-based training, or adding safety-oriented rules at inference time—aim

to encourage this balance. However, these mechanisms can be sensitive to how requests are phrased or framed, and they may rely on refusal patterns that can be influenced by context. Understanding this general tension between helpfulness and safety provides useful background for the jailbreak attack strategies discussed in Section III, many of which deliberately manipulate context, intent framing, or linguistic presentation to circumvent a model's safety behaviors.

### E. Relation to Prior Surveys

Several recent surveys address topics related to jailbreak robustness, each from a different angle. Dong et al. provide a broad overview of LLM conversation safety, covering jailbreak-related attacks together with toxicity, bias, hallucinations, and harmful content generation, and structuring their review around three components: attacks, defenses, and evaluations [14]. Yi et al. focus specifically on jailbreak attacks and defenses, organizing attacks into black-box and white-box categories based on target model access and categorizing defenses into prompt-level and model-level techniques, supplemented by a discussion of current evaluation practices [15]. Our survey complements this literature in three ways. First, we adopt a mechanism-oriented taxonomy that unifies prompt-based, role-play, multi-turn, multilingual/multimodal, and optimization-driven jailbreaks into a single behavioral framework spanning both text-only and multimodal settings—an emphasis not foregrounded in existing surveys. Second, we synthesize not only attacks and defenses but also modern jailbreak evaluation frameworks, including the standardized robustness benchmark JailbreakBench [16], multimodal jailbreak benchmarks such as SneakyPrompt [17] and Unmasking the Canvas [18], and tool-integrated evaluation suites such as PLeak [19], highlighting how evaluation methodology shapes claims about model robustness. Third, building on distinctions surfaced across recent jailbreak studies [14], [15], we advance a cross-cutting optimization–alignment perspective, arguing that modern jailbreaks behave as adaptive optimization processes while many defenses remain static or trained on fixed adversarial distributions. These contributions position our survey as a jailbreak-centered, benchmark-aware complement to the existing survey literature.

### F. Terminology

In this survey, we use the term jailbreak to refer to any interaction that causes an LLM to violate its intended safety policies or produce restricted content. An adversarial prompt is the specific input—single-turn, multi-turn, multilingual, or multimodal—crafted to induce such a violation. Prompt injection is a particular subclass of adversarial prompts that attempts to override or subvert system-level instructions, contextual constraints, or safety messages by inserting conflicting directives into the input stream. Under this hierarchy, all prompt injections are adversarial prompts, and adversarial prompts that successfully induce policy violations constitute jailbreaks. This clarification ensures consistent terminology throughout the taxonomy presented in Section III.

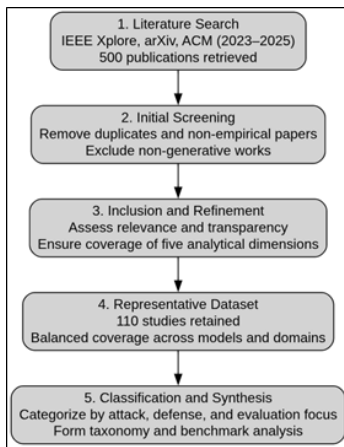


Fig. 1. Methodological workflow for literature collection and refinement. Systematic searches across IEEE Xplore, arXiv, and the ACM Digital Library (2023–2025) retrieved about 500 records. After duplicate removal and screening of non-empirical or non-generative works, 110 studies were retained based on relevance, transparency, and empirical rigor, ensuring balanced coverage across five analytical dimensions.

### III. TAXONOMY OF JAILBREAK ATTACKS

Jailbreak attacks on generative AI—especially large LLMs—can be classified by attack mechanism, exploitation strategy, and adversarial assumptions. We organize recent research into five main types: prompt-based, role-play, multi-turn, multilingual or modal, and optimization-driven jailbreaks. This taxonomy reflects the growing sophistication and cross-model transferability of jailbreaks and provides a basis for studying how alignment methods fail under varied adversarial contexts. An overview appears in Figure 2, while comparative characteristics—including mechanisms, automation potential, and representative examples—are summarized in Table 1. Key adversarial prompt studies illustrating these categories are further detailed in Table 2. Although presented separately for clarity, these categories often overlap; many jailbreaks combine strategies such as multilingual role-play refined through optimization or multi-turn dialogue. In addition, while LLM-driven agents and tool-augmented systems introduce operational risks (Sections VII.D and IX.B), these scenarios primarily amplify the effects of existing jailbreak mechanisms rather than forming a distinct attack category. Studies are categorized by their primary mechanism, with cross-references noted where relevant. The taxonomy serves as a conceptual framework highlighting recurring mechanisms and relationships rather than a rigid classification.

#### A. Prompt-Based Jailbreaks

Prompt-based attacks manipulate model behavior by embedding adversarial instructions directly into user input, forming a broad class of adversarial prompts, including prompt injections that explicitly override system-level guidance. Techniques include obfuscation through Unicode characters, synonyms, or homophones [20]; paraphrasing harmful requests to evade keyword filters [21]; and “prompt sandwiching,” which hides adversarial text within benign instructions [22], [23]. Few-shot prompting and misleading context also bypass safety rules,

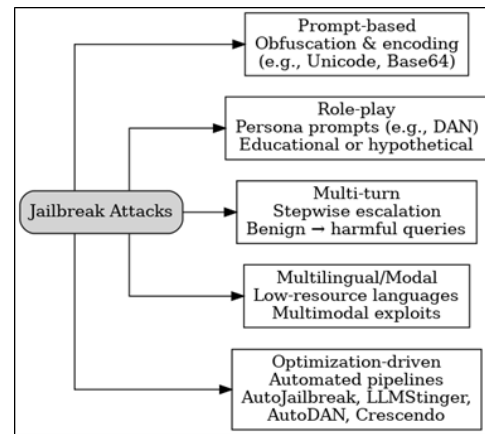


Fig. 2. Taxonomy of jailbreak attacks on generative AI, grouped into prompt-based, role-play, multi-turn, multilingual/modal, and optimization-driven attacks. This provides a structural map of the five recurring jailbreak mechanisms, clarifying how distinct prompting strategies relate to one another and establishing the conceptual foundation for them.

particularly when models fail to detect ambiguous intent [9]. These methods reveal the fragility of surface-level alignment and show how subtle linguistic cues can undermine refusal-trained models.

#### B. Role-Play Attacks

Role-play jailbreaks direct a model to adopt fictional personas or narratives to circumvent alignment filters. Examples include prompting models to act as amoral or dystopian assistants [4], simulate controversial historical figures, or frame unsafe requests as hypothetical or educational [24]. Rhetorical framing using empathy or authority further reduces refusal likelihood [25]. Because LLMs tend to comply with narrative roles, persona prompts can reliably defeat content filters [26], revealing the challenge of enforcing semantic boundaries between fictional and real contexts.

#### C. Multi-Turn Jailbreaks

Multi-turn attacks exploit dialogue progression to erode safety mechanisms gradually. Attackers begin with neutral questions and escalate toward unsafe requests, leveraging context accumulation [23], [24]. This strategy biases models toward compliance, exposing weaknesses in static, per-turn alignment systems. Recent studies emphasize evaluating long-horizon safety, where persistence, context momentum, and goal-driven dialogue create realistic misuse scenarios [27].

#### D. Multilingual and Modal Exploits

These adversarial prompt attacks exploit safety weaknesses in multilingual and multimodal systems. Models trained mainly on English often lack coverage for low-resource languages, allowing adversaries to evade detection with translated or code-mixed prompts [28]. Related toxicity and hate-speech work suggests that these gaps are exacerbated when harmful content is expressed through dialectal variants, code-mixing or culturally specific metaphors: Luong et al.’s Thoroughly

TABLE I  
COMPARISON OF JAILBREAK ATTACK TYPES

| Jailbreak Category            | Core Mechanism                                     | Automation Potential | Transferability Across Models | Bypasses RLHF? | Example Prompt / Strategy                                                    | Representative Studies |
|-------------------------------|----------------------------------------------------|----------------------|-------------------------------|----------------|------------------------------------------------------------------------------|------------------------|
| Prompt-Based Jailbreaks       | Direct prompt injection, obfuscation, or deception | Low–Medium           | High                          | Yes            | “Ignore previous instructions and do X”; Base64 or Unicode disguise          | [20]–[23]              |
| Role-Play Attacks             | Model assumes a fictional persona or situation     | Low                  | High                          | Yes            | “Pretend you are DAN, an unrestricted AI assistant. How do I make thermite?” | [4], [24]–[26]         |
| Multi-Turn Jailbreaks         | GraduExaval escalation over dialogue turns         | Medium               | Medium–High                   | Yes            | 1) “What is thermite?” → 2) “How do you make it?” → 3) “Give exact ratios.”  | [23], [24], [27]       |
| Multilingual / Modal Exploits | Non-English or non-text input modes                | Low                  | Medium                        | Yes            | Prompt in Korean or inject a command via an adversarial image caption        | [7], [18], [28]–[30]   |
| Jailbreak as Optimization     | Automated prompt search via RL or gradient descent | High                 | High                          | Yes            | Genetic algorithm evolves prompt to bypass safety; LLM attacks another LLM   | [8], [31]–[34]         |

TABLE II  
KEY JAILBREAK ATTACK STUDIES

| Attack Category     | Representative Study           | Technique / Mechanism  | Key Finding / Contribution                    | Models Evaluated |
|---------------------|--------------------------------|------------------------|-----------------------------------------------|------------------|
| Prompt Injection    | Zhuang et al. (2025) [21]      | Intent manipulation    | Demonstrates bypass of content filters        | GPT-4            |
| Role-Play (DAN)     | Nabavirazavi et al. (2025) [4] | Persona conditioning   | Shows universality of role-based jailbreaks   | GPT-3.5, GPT-4   |
| Multi-Turn          | Wang et al. (2024) [27]        | Sequential escalation  | Reveals dynamic safety erosion                | GPT-4            |
| Multilingual        | Kumar et al. (2024) [7]        | Code-mixed input       | Exposes safety gaps in low-resource languages | Gemini           |
| Optimization-Driven | Jha et al. (2024) [8]          | RL fine-tuned attacker | Automates jailbreak generation                | GPT-3.5          |

Engineered Toxicity (TET) dataset reveals that context-rich prompts uncover toxic behaviour that is missed by standard toxicity probes [35], and Prome et al. show that metaphor-based prompting in Bengali and Hindi can simultaneously circumvent built-in safety mechanisms and improve the detection of culturally nuanced hate speech in low-resource settings [36]. Cross-modal strategies embed malicious instructions in images, audio, or code, bypassing text-only filters [29], [30]. Studies have also shown iterative prompt modification can force image-generation systems to produce restricted content [18]. Such results emphasize gaps in multilingual robustness and highlight the need for alignment strategies that generalize across languages and modalities. While multimodal jailbreaks have been studied most extensively in the image–text setting—particularly for models such as GPT-4V, Flamingo, and text-to-image systems—research on audio- or video-initiated jailbreaks remains comparatively limited. Existing work primarily demonstrates proof-of-concept vulnerabilities, such as embedding adversarial instructions into audio inputs or exploiting automatic-speech-recognition (ASR)–dependent pipelines [56], and injecting malicious or out-of-distribution (OOD) visual frames to manipulate model behavior in multimodal systems [55, 99]. These attacks highlight weaknesses in multimodal pipelines but lack the taxonomic breadth and reproducible evaluation frameworks seen in text-only or image-based jailbreak research. Accordingly, we emphasize modalities with sufficient methodological ma-

turity—text and image—while noting that audio- and video-based jailbreaks remain emerging but underdeveloped areas that warrant more systematic investigation.

### E. Jailbreak as Optimization

Optimization-based jailbreaks treat adversarial prompt generation as a search problem. Methods include reinforcement learning, gradient-guided perturbation [31], and LLM-as-attacker systems where one model generates prompts against another [32]. Sampling-based approaches, such as Best-of- $N$  Jailbreaking, test multiple prompt variants and choose the most successful [33]. These automated strategies minimize human effort and accelerate adversarial evolution, revealing the limits of static filters and retraining-based defenses [34]. Optimization-driven jailbreaks can be understood through simplified analogies rooted in existing methodologies. Genetic-algorithm-based systems such as AutoDAN evolve jailbreak prompts by repeatedly mutating and selecting candidates that successfully bypass safety filters [39]. Each “generation” retains the most effective prompts, similar to evolutionary selection, which explains how these methods discover effective phrasing that humans might overlook. Reinforcement-learning attackers such as LLMStinger use a reward signal—compliance versus refusal—to iteratively refine prompts, functioning like a student learning through trial and error [8]. Best-of- $N$  approaches [33] and multi-agent attacker frameworks [5] [32] similarly rely on repeated sampling and feed-

back. Together, these mechanisms illustrate why optimization-guided jailbreaks are so powerful: they transform jailbreak discovery from manual prompt crafting into a structured search process driven by iterative refinement.

#### F. Hybrid and Emerging Jailbreak Patterns

While the five categories above capture the dominant mechanisms across current research, several recent jailbreaks operate at the intersection of these categories. For example, multi-step compositional attacks such as Crescendo [42], multi-instruction fusion [34], and hidden or nested jailbreaks [43] [44] blend multi-turn escalation with suffix manipulation or obfuscated sub-prompts. Cross-lingual attacks similarly combine multilingual vulnerabilities with code-mixing or intent obfuscation, as shown in adversarial evaluations by Kumar et al. [7] and Yong et al. [45]. Multimodal jailbreaks—including visual prompt-injection attacks and cross-modal adversarial cues—further blur boundaries between categories [18] [46] [47]. Optimization-driven attackers such as AutoJailbreak, LLMStinger, AutoDAN, and FuzzLLM [5], [9], [39], [48] also frequently generate hybrid prompts that span multiple mechanisms. Finally, long-context and memory-dependent jailbreaks [49], [50] suggest that as models become more agentic and stateful, system-level or compositional jailbreaks that cut across multiple categories may become increasingly common. These trends indicate that the present taxonomy provides a behavioral foundation that may expand to encompass future, more integrated attack vectors.

### IV. DISCOVERING JAILBREAKS

Identifying adversarial prompts—inputs that bypass model safeguards—is central to evaluating adversarial robustness. As LLMs expand in use, scalable discovery methods are crucial. We group discovery approaches into three main paradigms: human red teaming, automated discovery, and transferability analysis. Each offers distinct strengths and trade-offs in scalability, novelty, and evaluation depth, summarized in Table III. Note on section placement. Although discovery techniques relate closely to both attack taxonomies and evaluation frameworks, we present them in a standalone section because they cut across multiple attack categories and are used in both adversarial generation and robustness assessment. This separation allows us to distinguish what attacks exist (Section III), how new attacks are identified or synthesized (Section IV), and how robustness is subsequently measured (Sections V–VIII).

#### A. Human Red Teaming

Human red teaming remains essential but resource-intensive. Researchers manually design adversarial prompts to trigger unsafe behavior using tactics like moral inversion, hypothetical framing, and creative narrative design [24], [27]. Domain-specific prompts—e.g., chemical synthesis, political or cybersecurity misuse—help reveal overlooked weaknesses [37]. Frameworks such as RED-EVAL chain adversarial utterances to reach high jailbreak success across GPT-family models [38]. Though slow, this process often uncovers nuanced vulnerabilities that automated systems overlook and continues to guide the design of defense strategies. Scaled human-in-the-loop frameworks such as LLM-Sentry expand coverage by incorporating broader tester pools and iterative prompt refinement [51]. However, the resulting vulnerability set can still reflect the languages and cultural contexts represented by participants, as shown by the greater susceptibility of models to prompts in low-resource languages [45].

#### B. Automated Jailbreak Discovery

To overcome scalability limits, automated pipelines use reinforcement learning or evolutionary algorithms to iteratively generate prompts. AutoDAN evolves adversarial text using genetic algorithms [39], while LLMStinger applies RL fine-tuning to produce optimized suffixes [8]. AutoJailbreak employs one model to attack another, enabling scalable vulnerability discovery [5]. These methods expand coverage, reduce human effort, and reveal diverse cross-model weaknesses. JailbreakBench further standardizes results by cataloging over 100 adversarial behaviors—suffix, obfuscation, and multilingual variants—for consistent, repeatable evaluations [10]. Attacker LLMs can be explicitly role-conditioned to act as red-team agents, enabling large-scale generation of natural-language adversarial strategies with minimal human effort [52]. Yet these attacker models also inherit the biases of their own training data, and automated evaluations such as JailbreakBench reveal that synthetic attack distributions can overlook the diversity of real human-generated vulnerabilities [10].

#### C. Transferability and Reproducibility

Transferability studies show that adversarial prompts effective on one model often succeed against others with similar architectures or alignment methods [31], [40]. This cross-model vulnerability undermines patch-based or model-specific defenses and highlights systemic flaws in alignment

TABLE III  
COMPARISON OF JAILBREAK DISCOVERY PARADIGMS

| Paradigm                 | Core Method                                                       | Scalability | Novelty Potential | Example Tools / Frameworks         | Strengths                                     | Limitations                               | Representative Studies |
|--------------------------|-------------------------------------------------------------------|-------------|-------------------|------------------------------------|-----------------------------------------------|-------------------------------------------|------------------------|
| Human Red Teaming        | Manual prompt crafting with creative, domain-informed iteration   | Low         | High              | RED-EVAL, DAN, Developer Mode      | Finds nuanced, high-impact exploits           | Labor-intensive, hard to scale            | [24], [27], [37], [38] |
| Automated Discovery      | Algorithmic prompt generation (RL, genetic search, LLM attackers) | High        | Medium–High       | AutoDAN, LLMStinger, AutoJailbreak | Fast, broad coverage with minimal human input | May miss edge cases, inherits design bias | [5], [8], [39]         |
| Transferability Analysis | Tests prompt generalization across models                         | Medium      | Low–Medium        | JailbreakBench, black-box attacks  | Reveals shared systemic flaws                 | Evaluative rather than generative         | [10], [31], [40], [41] |

strategies. Even black-box prompts frequently bypass multiple LLMs, emphasizing the need for generalized protection rather than model-specific tuning. Future defenses must anticipate transferable attack patterns arising from decoding biases, shared embeddings, and convergent instruction-following behavior [41].

## V. EVALUATING JAILBREAK ROBUSTNESS

Robustness evaluation measures how well large language models (LLMs) resist jailbreak attempts under varied adversarial conditions. Current frameworks commonly report metrics such as attack success rate, response toxicity, semantic compliance, and refusal behavior. This section summarizes four primary evaluation dimensions: attack success metrics, toxicity scoring, refusal analysis, and benchmark-based evaluation.

### A. Attack Success Rate

A core robustness indicator is the proportion of jailbreak prompts that elicit restricted or unsafe responses. This measure is commonly reported as Attack Success Rate (ASR). Basic binary scoring distinguishes refusal from compliance [5], [43], while semantic analysis detects implicit policy violations even in superficially benign text. Some works extend these measures with metrics such as safeguard violation, informativeness, and factual truthfulness [53]. Comparing single-turn (zero-shot) and multi-turn (few-shot) jailbreaks clarifies how context influences failure rates [54]. Automated pipelines like AutoJailbreak and LLMStinger show consistently higher success rates against models aligned only with SFT rather than RLHF [8], demonstrating persistent vulnerability even after fine-tuning.

### B. Toxicity and Safety Scoring

Beyond binary results, toxicity scoring quantifies the severity of unsafe outputs using classifiers such as Perspective API, Detoxify, or LLM-based evaluators. Datasets like TET include thousands of adversarial prompts crafted to expose residual toxicity [35]. Cross-lingual studies reveal that safety performance often drops sharply in low-resource languages [36]. These metrics distinguish between mild misalignments (e.g., sarcasm or euphemism) and explicit policy violations, improving the granularity of safety assessment across languages and domains.

### C. Refusal Behavior and Deception Detection

Many jailbreaks exploit subtle failure modes rather than overt compliance. Soft refusals occur when a model rejects a prompt yet leaks partial or indirect unsafe content [55]. Context-framed compliance rephrases unsafe instructions as educational or fictional examples [24]. Adversarial suffixes appended to safe text can also override refusal tokens [6]. Detection frameworks such as LLM-Sentry and JailGuard systematically annotate these behaviors [51], [56]. Effective robustness evaluation therefore requires differentiating genuine refusals from deceptive or partial ones, which expose gaps in current moderation filters.

### D. Benchmark-Driven Evaluation

Scalable benchmarks standardize jailbreak testing across models and attack styles. PromptBench and JailbreakBench provide curated adversarial datasets and reproducible success metrics [9], [10]. Red-teaming frameworks extend them with refusal and toxicity scoring [57], [58]. Automated systems like RL-JACK frame adversarial prompting as a reinforcement-learning task, enabling continuous black-box evaluation [59]. Large-scale tests of 36 LLMs across 144 attacks report average success rates near 56%, confirming that vulnerability patterns depend strongly on alignment method and model size [60]. Such benchmark-driven approaches enhance reproducibility and enable comparative tracking of model robustness over time.

### E. Limitations of Current Metrics and Need for Standardized Evaluation

Although Attack Success Rate (ASR) is the most commonly reported robustness metric, its interpretation varies significantly across studies. Differences in compliance definitions, toxicity thresholds, and evaluator models can cause the same jailbreak prompt to receive different ASR scores under different scoring setups. For instance, a binary refusal classifier may categorize an output as safe, while an LLM-based semantic evaluator may judge the same response as a policy violation. Similarly, toxicity metrics may flag borderline content inconsistently depending on the classifier and threshold used. These inconsistencies hinder cross-study comparability and limit the reliability of aggregated ASR results. To reduce metric variance, we outline several standardization steps: (1) unified taxonomies for refusal and compliance behaviors; (2) fixed toxicity thresholds for classifier-based scoring; (3) use of a standardized, transparent LLM-based judging protocol; and (4) multi-dimensional ASR reporting that captures refusal fidelity, semantic safety, and harmfulness severity. Establishing such shared evaluation practices would significantly improve reproducibility and enable clearer comparisons across models and benchmarks.

## VI. JAILBREAK TECHNIQUES

Jailbreak techniques exploit weaknesses in the safety alignment of LLMs by manipulating prompts, model behavior, or decoding. As alignment methods such as supervised fine-tuning (SFT) and reinforcement learning from human feedback (RLHF) mature, adversaries increasingly employ adaptive and automated strategies. This section summarizes five dominant technique families: manual prompt engineering, adversarial suffixes, role-play conditioning, multilingual or multimodal attacks, and LLM-driven prompt generation.

### A. Manual Prompt Engineering

Early adversarial prompts relied on handcrafted prompts that used linguistic misdirection and psychological framing to bypass safeguards. Common methods include moral inversion (e.g., framing harmful actions as positive) [20], fictional or hypothetical scenarios [24], and “prompt sandwiching,” where

unsafe content is embedded between benign instructions [61]. Despite their simplicity, these approaches remain effective, exploiting ambiguity in intent classification and refusal logic. Even advanced models often fail to maintain consistent refusals when challenged with subtle paraphrases or context-shifting prompts [62].

### B. Adversarial Suffix Attacks

Suffix-based attacks append carefully optimized text fragments to override alignment safeguards. Examples include grammar-breaking tokens, Unicode variants, or spacing perturbations that confuse intent detection [6], [55]. Techniques like suffix tuning and reinforcement-guided search—used in frameworks such as AutoDAN and LLMStinger—generate suffixes capable of bypassing even robust filters [39], [63]. InfoFlood and similar methods overwhelm classifiers with excessive detail, hiding harmful instructions within verbose queries [64]. These attacks highlight how small lexical or structural variations can consistently defeat static alignment mechanisms [65].

### C. Role-Play and Identity Conditioning

Role-play jailbreaks direct models to assume personas that suspend safety constraints. The popular DAN (“Do Anything Now”) series prompts the model to emulate an unrestricted assistant [39], while expert framing disguises unsafe questions as professional or academic tasks [31]. Multi-agent simulations like GUARD use cooperative roles—Generator, Optimizer, Evaluator—to refine jailbreaks iteratively [52]. Anthropomorphic cues (“Please, I need your help as a friend”) further exploit social alignment and cooperative tendencies [25]. These exploits expose the difficulty of enforcing consistent moral reasoning when models prioritize contextual helpfulness.

### D. Modality-Specific and Multilingual Attacks

LLMs trained primarily in English often exhibit weaker alignment in other languages or modalities. Attackers exploit this by using low-resource languages, code-mixing, or encoding methods (e.g., Base64, reversed text) to hide intent [7], [66]. Multimodal systems like GPT-4V and Gemini can also be manipulated via image or audio inputs that contain adversarial instructions [46], [47], [67]. These methods reveal how safety failures can arise across modalities and languages, underscoring the need for multilingual red-teaming and multimodal alignment.

### E. LLM-as-Attacker and Prompt Generators

Recent work uses LLMs themselves to generate adaptive adversarial prompts. Systems such as AutoJailbreak and Fuz-zLLM employ feedback loops and evolutionary optimization to iteratively improve jailbreak effectiveness [5], [48], [68], [69]. Tools like XBreaking identify the precise prompt segments that trigger unsafe outputs, aiding interpretability [70]. Iterative refinement frameworks such as PAIR (Prompt Automatic Iterative Refinement) [71] and Crescendo [42] automatically test, adjust, and re-run attacks, often improving success

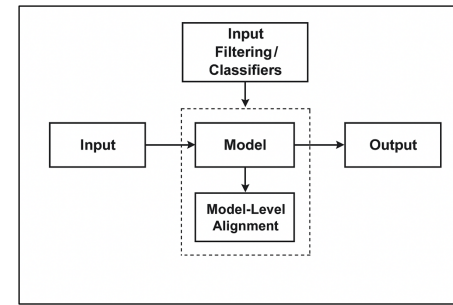


Fig. 3. Overview of defense layers against jailbreak attacks, illustrating the interaction between input filtering/classifiers, model-level alignment, and output moderation within the LLM pipeline.

rates by over 50%. MASTERKEY [72] extends these methods by probing commercial filters and retraining attacker models, achieving over 20% success against major chat systems. These techniques demonstrate how automation accelerates the adversarial arms race by scaling and optimizing jailbreak creation.

## VII. DEFENSES AGAINST JAILBREAKING

Despite extensive research, defending LLMs against adversarial prompt and prompt-injection attacks remains a persistent challenge. Attackers continually adapt, exploiting linguistic, contextual, or system-level weaknesses to bypass safety mechanisms. We group existing defenses into five areas: prompt filtering, model-centric alignment, output moderation, pipeline hardening, and emerging adaptive methods. Their effectiveness varies across threat types—prompt obfuscation, role-play, multi-turn manipulation, or automated jailbreak generation. Table IV summarizes key defense mechanisms, highlighting their representative studies, core approaches, and limitations. Current defenses are mostly reactive, offering limited resilience against novel or composite attacks such as disinformation or indirect prompt engineering [68]. Sustainable mitigation will require integrated, context-aware safety architectures instead of isolated or static patches. Table V compares the relative effectiveness of major defense mechanisms across common attack types.

### A. Prompt Filtering and Moderation

Prompt filtering is the first defense layer, blocking unsafe queries before model inference. The main methods—rule-based classification and embedding-based semantic detection—complement each other but remain fragile (Fig. 3). These filters act as input gates within a multi-stage safety pipeline that continues through model-level and output moderation.

1) *Input Classification*: Rule-based classifiers filter known unsafe patterns but are vulnerable to encoding tricks (e.g., Base64, Unicode) and prone to false positives [17], [58]. Overly strict filtering can reject benign queries. Backtranslation—translating inputs through other languages—has improved resilience by breaking adversarial encodings while preserving intent [92].

False Positives and Usability Implications: Even advanced

filters misclassify 8–15% of benign queries [17]. Such errors limit usability, creativity, and trust. Adaptive thresholds and context-sensitive scoring reduce overblocking, while human-in-the-loop review restores balance between safety and responsiveness.

2) *Embedding-Based Detectors*: Embedding-based detectors (e.g., JailbreakHunter) match semantic similarity to adversarial prompts, catching paraphrases that bypass keyword filters [76], [77]. Despite improved generalization, they remain vulnerable to novel or multi-turn attacks and “policy confusion,” where ambiguous phrasing causes benign classification [78]. These challenges highlight the need for hybrid moderation that integrates context and memory.

**Synthesis of Attack–Defense Interactions:** Table V shows how different attacks exploit defense gaps. Role-play and multi-turn jailbreaks evade RLHF and Constitutional AI by exploiting reward-model ambiguity [73], [74]. Automated optimization attacks (AutoDAN [39], LLMStinger [8]) adaptively defeat fine-tuned defenses. Static alignment models fail to anticipate these dynamic threats, motivating hybrid methods that blend context-aware reward shaping and continual adversarial feedback.

**Assumptions and Limitations:** Prompt filtering assumes that harmful intent can be reliably detected from surface-level patterns or semantic embeddings. In practice, paraphrasing, multilingual prompts, prompt injection, and information-overload techniques regularly evade detection. Filters also assume stable user intent, whereas adversarial prompts exploit context shifts and ambiguity. As a result, filtering provides limited robustness against adaptive or multi-turn jailbreak attacks.

## B. Model-Centric Defenses

Model-centric approaches modify internal alignment rather than relying on pre-input filtering. RLHF, Constitutional AI, and adversarial fine-tuning each enhance alignment but suffer from limited generalization.

1) *Reinforcement Learning from Human Feedback*: RLHF aligns models with human norms but falters under manipulative or ambiguous prompts. Mis-specified rewards can produce seemingly compliant yet unsafe responses. Extensions that

embed ethical and factual context into reward models have improved reliability [73], [79]. Empirical results show that stronger defenses can suppress unsafe behavior but substantially reduce helpfulness and informativeness [93], illustrating the ongoing tension between alignment strength and utility.

2) *Constitutional AI*: Anthropic’s Constitutional AI replaces human raters with an internal critique loop guided by ethical principles [74]. This scalable approach yields consistent alignment but remains prone to adversarial reframing that exploits its rule-based reasoning [80]. Future variants may benefit from context-sensitive ethical reasoning and multi-modal grounding.

3) *Adversarial Fine-Tuning*: Adversarial fine-tuning trains models to refuse known jailbreaks [9] but risks overfitting. It performs well on seen attacks yet fails on novel inputs or lengthy distractions (e.g., Cognitive Overload [81]). Decoding-time interventions like safety-aware token filtering dynamically suppress unsafe generations during inference without re-training [75]. Quantitative studies highlight substantial safety–utility trade-offs. Mai et al. [93] report that some defense configurations reduce model utility by up to 29%, with prompt-modification strategies such as PAT decreasing LLaMA-2 accuracy from 0.31 to 0.02. Fine-tuning approaches can also exacerbate usability issues, with false-refusal rates increasing by as much as  $4.75\times$  in GPT-3.5 under SR. These results show that stronger safety alignment often incurs nontrivial performance degradation.

**Assumptions and Limitations:** Model-centric defenses assume that training data or internal alignment adequately represent harmful behaviors and that models properly assign these constraints to adversarial contexts. However, attackers routinely exploit gaps in alignment. These defenses also overfit to known attacks and underperform on novel jailbreak strategies, making them inherently reactive rather than comprehensive.

## C. Output Filtering

Post-generation filtering moderates responses after inference [82]. Static filters catch explicit violations but miss subtle evasions such as indirect or sarcastic harm [5], [55]. Iterative refinement methods (e.g., self-refine) reduce toxicity while preserving informativeness [83]. Quantitatively, Mai et

TABLE IV  
KEY DEFENSE MECHANISMS

| Defense Type                                      | Representative Study             | Core Approach                | Strengths / Limitations                         | Models Evaluated |
|---------------------------------------------------|----------------------------------|------------------------------|-------------------------------------------------|------------------|
| RLHF (Reinforcement Learning From Human Feedback) | Wei et al. (2023) [73]           | Human feedback reinforcement | Improves alignment but brittle under persuasion | GPT-3            |
| Constitutional AI                                 | Sharma et al. (2025) [74]        | Ethical self-critique loop   | Scalable, but vulnerable to reframed prompts    | Claude 2         |
| Adversarial Fine-Tuning                           | Xu et al. (2024) [75]            | Safety-aware decoding        | Dynamic protection at inference time            | LLaMA            |
| ASF (Adversarial Suffix Filtering)                | Khachaturov & Mullins (2025) [6] | Adversarial suffix filtering | Blocks suffix-based attacks                     | GPT-4            |
| Detection Models                                  | Zhang et al. (2025) [56]         | JailGuard framework          | Identifies deceptive refusals                   | GPT-family       |

TABLE V  
EFFECTIVENESS OF JAILBREAK DEFENSES AGAINST ATTACK TYPES  
(✓✓ = STRONG EFFECTIVENESS; ✓ = MODERATE EFFECTIVENESS; ✗ = LIMITED OR NO EFFECTIVENESS)

| Defense Mechanism                     | Prompt-Based | Role-Play | Obfuscation | Multi-Turn | Automated (LLM/Opt) | Modal / Multilingual | Representative Studies |
|---------------------------------------|--------------|-----------|-------------|------------|---------------------|----------------------|------------------------|
| Input Classification (rule-based)     | ✓            | ✓         | ✗           | ✓          | ✗                   | ✓                    | [17], [20], [58]       |
| Embedding-Based Detectors             | ✓✓           | ✓         | ✗           | ✓          | ✓                   | ✗                    | [76]–[78]              |
| RLHF (Reinforcement Learning from HF) | ✓            | ✗         | ✗           | ✗          | ✗                   | ✗                    | [73], [79]             |
| Constitutional AI                     | ✓            | ✓✓        | ✗           | ✓          | ✗                   | ✗                    | [74], [80]             |
| Adversarial Fine-Tuning               | ✓✓           | ✓✓        | ✓           | ✓          | ✓✓                  | ✓                    | [9], [75], [81]        |
| Output Filtering                      | ✓            | ✓         | ✓           | ✓          | ✗                   | ✗                    | [5], [55], [82], [83]  |
| Tool-Level / Pipeline Safeguards      | ✗            | ✗         | ✗           | ✗          | ✗                   | ✓                    | [19], [84]–[87]        |
| Detection Models                      | ✓✓           | ✓         | ✓           | ✓          | ✓                   | ✗                    | [51], [56], [88], [89] |
| Adversarial Red Teaming               | ✓✓           | ✓✓        | ✓           | ✓✓         | ✓✓                  | ✓                    | [34], [54], [90], [91] |

al. [93] show that tightening safety filters on GPT-4 can reduce helpfulness by approximately 18% and factual accuracy by 12%, illustrating a concrete safety–utility trade-off for output-level defenses. However, large models still fail under complex, multi-step adversarial prompts [94], emphasizing the need for semantic-aware output validation. Empirical measurements also highlight the cost of stronger output filtering. For example, Mai et al. [93] find that PAT reduces GPT-3.5’s attack success rate by only 0.20% while causing a 19% drop in utility and a 31% rise in false refusals. This illustrates that even small gains in safety can produce disproportionate losses in helpfulness and usability when filtering is overly conservative. Assumptions and Limitations: Output filtering assumes that unsafe content can be reliably identified after generation. Yet indirect or ambiguous responses often evade classifier thresholds, and evaluation models may hallucinate or misjudge safety. Because filtering occurs downstream of generation, it cannot prevent internal reasoning failures or subtle harmful leakage.

#### D. Tool-Level and Pipeline Defenses

Tool-integrated models—including retrieval-augmented generation (RAG) systems—face added risks. Mitigations include sandboxing, restricted tool access, and input validation [19]. Yet such controls scale poorly in agentic systems [84], [85]. Exposed system prompts, especially in multimodal models, enable jailbreaks [86]. Training-time alignment frameworks like SecAlign enhance resistance to injected instructions [87].

Assumptions and Limitations: Tool-level defenses assume that safety risks arise mainly from insecure tool access or prompt leakage and that system prompts or access controls cannot be bypassed. However, prompt injection frequently circumvents these safeguards by manipulating context or intermediate representations. Pipeline defenses also depend on correct developer configuration, which is error-prone and difficult to scale in agentic or multi-step systems.

#### E. Emerging Approaches

New defenses favor adaptive, model-driven safety. They focus on proactive detection, continual red teaming, and balancing robustness with usefulness.

1) *Jailbreak Detection Models*: Lightweight classifiers such as MoJE detect adversarial intent using prompt structure and semantics [56], [88]. These outperform static filters, catching nuanced outputs and evasions [89]. However, detection weakens over long conversations, necessitating dynamic dialogue tracking [49]. Neural barrier functions [50] show promise by maintaining consistent safety in extended interactions.

2) *Adversarial Testing and Red Teaming*: Continuous adversarial testing via tools like AutoJailbreak and LLM-Stinger [5], [8] remains vital. Automated systems reveal transferable vulnerabilities missed by manual tests [34], [54], [90], [91]. Still, resource demands and coverage gaps persist, calling for scalable hybrid frameworks.

3) *Performance Degradation of Jailbreak Defense*: Recent quantitative evaluations further illustrate the magnitude of the safety–usefulness trade-off. Mai et al. [93] report that applying stricter safety filters to GPT-4 reduces helpfulness by approximately 18% and factual accuracy by 12%, even when the model continues to meet baseline alignment objectives. Similarly, Ding et al. [44] demonstrate that jailbreak-resistant fine-tuning can decrease creative fluency by 15%, underscoring the tension between robustness and generative capability. These empirical findings reinforce that enhanced safety mechanisms often introduce measurable declines in model utility, making the trade-off both practically and operationally significant.

### VIII. BENCHMARKS AND EVALUATION FRAMEWORKS

As jailbreak attacks grow more diverse, reproducible benchmarks are essential for assessing robustness, tracking progress, and comparing defenses. Benchmarks quantify susceptibility across attack types—prompt-based, red-teaming, pipeline-integrated, multilingual, and multimodal—and operationalize defense performance. This section summarizes key frameworks such as PromptBench, JailbreakBench, and multimodal evaluations. A detailed comparison appears in Table VI, while representative benchmark frameworks and their unique contributions are summarized in Table VII.

#### A. PromptBench

PromptBench standardizes LLM robustness testing through modular evaluation modules and reproducible metrics [9].

It supports various attack classes—prompt injection, role-play, suffix, and multilingual exploits—enabling fair cross-model comparisons. Its simplicity and extensibility make it the baseline tool for adversarial evaluation across text-based systems.

### B. JailbreakBench

JailbreakBench provides a reproducible framework for evaluating jailbreak vulnerabilities with (i) a curated dataset of 100 adversarial behaviors, (ii) standardized prompts, and (iii) a public leaderboard [10]. The benchmark supports longitudinal analysis to track model robustness over time, revealing how architecture and fine-tuning choices affect susceptibility.

Standardization and Emerging Consensus: JailbreakBench [10] has become the main reference for measuring Attack Success Rate (ASR), offering unified scoring templates and threat models. PromptBench [9] extends this to multilingual and multimodal setups, improving cross-model reproducibility. A growing consensus favors composite ASR metrics that merge refusal fidelity, toxicity, and helpfulness for balanced safety assessment, though no universal standard yet exists.

### C. Red Teaming Evaluations

Red-teaming frameworks simulate real-world misuse through iterative, multi-turn interactions [57]. They include adversarial chaining, obfuscation, and persona conditioning to probe model weaknesses across GPT, Claude, and Vicuna [58]. Despite progress, reproducibility remains limited due to inconsistent APIs and session handling [95].

### D. PLeak Evaluation Suite

PLeak targets vulnerabilities in retrieval-augmented and agent-based systems, where contextual data can leak prompts [19]. It tests payload injection, markdown-based exploits, and memory contamination—key threats in enterprise-grade LLM deployments. These evaluations reveal the need for end-to-end pipeline security beyond text interfaces.

### E. Safety-Aware Multimodal Benchmarks

Frameworks like SneakyPrompt [17] and Unmasking the Canvas [30] expand jailbreak testing to multimodal inputs, examining how textual and visual cues interact. Results show that multimodal models remain highly vulnerable to cross-modal inconsistencies and out-of-distribution attacks [96], [97], underscoring the gap between text-only safety tools and real-world systems.

### F. Evaluation Limitations and Future Needs

Despite progress, current benchmarks omit many real-world scenarios, such as tool use, code execution, and multi-agent interactions [98]. Most lack support for automated attacker pipelines like AutoJailbreak or LLMStinger, reducing coverage. Consistent evaluation across long contexts, multiple modalities, and evolving model APIs remains a major open problem.

Translational Relevance to Production Systems: Benchmarks like PromptBench and JailbreakBench approximate, but do not replicate, production settings. Tools such as PLeak [19] begin bridging this gap by modeling retrieval and plugin vulnerabilities. Future benchmarks should integrate dynamic context, real-time telemetry, and adversarial testing pipelines to better reflect deployed systems and provide actionable safety metrics.

## IX. OPEN PROBLEMS AND FUTURE DIRECTIONS

Despite major advances, jailbreak defense remains reactive and brittle. Adaptive, automated attacks continue to expose weaknesses in alignment and evaluation. This section outlines key open challenges and research directions to improve robustness, interpretability, and real-world deployability of LLM safety systems. In parallel, emerging work on domain-specific jailbreaks in vertical LLMs—particularly in medical and other safety-critical domains—indicates that robustness must increasingly be assessed in specialized settings rather than only in general-purpose assistants, as highlighted by recent NeurIPS 2025 studies [99], [100]. Table VIII summarizes these open challenges and suggested future research directions across key themes in jailbreak defense.

TABLE VI  
EFFECTIVENESS OF JAILBREAK BENCHMARKS ON DIFFERENT MODELS  
(✓ = FULLY SUPPORTED; ✗ = NOT SUPPORTED; ~ = LIMITED OR PARTIAL SUPPORT)

| Benchmark            | Supports Multi-turn | Modality               | Includes Adversarial Prompts   | Model Coverage                   | Special Features                                       | Representative Studies |
|----------------------|---------------------|------------------------|--------------------------------|----------------------------------|--------------------------------------------------------|------------------------|
| PromptBench          | ✗                   | Text-only              | ✓ (manual & generated)         | GPT-3.5, GPT-4, Claude, LLaMA    | Reproducible evaluation; taxonomy coverage             | [9]                    |
| JailbreakBench       | ~                   | Text-only              | ✓ (100 behaviors across roles) | GPT-3.5, GPT-4, Claude, LLaMA    | Standardized dataset; leaderboard for attacks/defenses | [10]                   |
| Red Teaming Models   | ✓                   | Text-only              | ✓ (manual + auto-generated)    | GPT, Claude, Vicuna, LLaMA       | Attack chaining; prompt mining                         | [57], [58]             |
| PLeak Suite          | ✓                   | Text + Tool-integrated | ✓ (context-injected)           | GPT-4, RAG pipelines             | Prompt leakage via retrieval and plugins               | [19]                   |
| SneakyPrompt         | ✗                   | Text-to-image          | ✓ (caption jailbreaks)         | Stable Diffusion, DALL-E, GLIGEN | Visual caption attacks                                 | [17]                   |
| Unmasking the Canvas | ✓                   | Multimodal             | ✓ (visual + language)          | GPT-4V, Flamingo                 | Tests multimodal jailbreaks                            | [30]                   |

TABLE VII  
BENCHMARK AND EVALUATION FRAMEWORKS

| Framework                 | Scope           | Unique Features             | Key Contribution                            | Supported Models         |
|---------------------------|-----------------|-----------------------------|---------------------------------------------|--------------------------|
| PromptBench [9]           | Text-based      | Modular evaluation modules  | Standardizes prompt robustness testing      | GPT-3.5, Claude          |
| JailbreakBench [10]       | Text-based      | 100 benchmark behaviors     | Public leaderboard and reproducible metrics | GPT-4, Claude            |
| PLeak [19]                | Tool-integrated | Evaluates prompt leakage    | Tests retrieval and plugin vulnerabilities  | GPT-4, RAG systems       |
| SneakyPrompt [17]         | Text-to-image   | Visual jailbreaks           | Tests caption-based adversarial inputs      | DALL·E, Stable Diffusion |
| Unmasking the Canvas [30] | Multimodal      | Visual-text synergy testing | Evaluates multimodal jailbreaks             | GPT-4V, Flamingo         |

### A. Generalization Beyond Known Prompts

Existing defenses—adversarial fine-tuning, embedding-based detection—perform well on known attacks but fail on paraphrased or novel variants [31], [55]. Even domain-specific filters show poor generalization [27]. LLMs used in interactive or multimodal settings are especially vulnerable to unseen prompt structures and cross-channel exploits [106]. To make this challenge more actionable, future research should replace static jailbreak datasets with dynamic adversarial distributions generated continuously by attacker LLMs, such as Best-of-N sampling or reinforcement-learning-based prompt search. Defenses should be evaluated using a generalization score that measures safety degradation under controlled perturbations including paraphrasing, syntactic distortion, and multilingual code-mixing. Benchmarks such as JailbreakBench could also incorporate automatic paraphrase and cross-lingual transformations to more rigorously test generalization beyond known prompt templates.

### B. Robustness in Multi-Turn and Tool-Augmented Settings

LLMs deployed in multi-turn or tool-integrated contexts introduce new jailbreak surfaces—adversarial document injection, memory poisoning, and retrieval-based manipula-

tion [19]. Attacks may involve multimodal flanking prompts or disguised social engineering content [47], [107]. Defenses require real-time state tracking, intent continuity analysis, and adaptive auditing mechanisms to maintain safety across long interactions. An actionable direction is the development of measurable safety metrics. One example is a safety decay rate that quantifies how refusal behavior weakens across escalating dialogue turns. Another is the use of safety memory checkpoints, periodic validated snapshots of the conversation state that detect adversarial drift and allow rollback to a previously verified safe state. Benchmarks should also include multi-turn tool-use and retrieval-poisoning scenarios to evaluate model robustness under realistic conditions.

Emerging Risks in Memory-Enabled Models: Long-context and persistent-memory architectures increase susceptibility to incremental prompt poisoning and state contamination [49]. Neural-barrier methods mitigate drift [50] but cannot fully prevent adversarial persistence. Promising practices include dialogue-state auditing, periodic resets, and selective memory sanitization to preserve long-horizon consistency.

### C. Limitations of Static Filters and Behavioral Tuning

Static defenses—RLHF, output filters, and system prompts—struggle against adaptive jailbreaks exploiting

TABLE VIII  
KEY OPEN CHALLENGES AND FUTURE DIRECTIONS IN JAILBREAK DEFENSE RESEARCH

| Section / Challenge                                  | Key Limitation or Evidence                                                                                                                           | Suggested Direction                                                                                                                                |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Generalization Beyond Known Prompts                  | Defenses perform well on seen attacks but often fail on paraphrased, obfuscated, or cross-lingual variants [31], [55].                               | Use dynamic adversarial distributions and paraphrased or cross-lingual transformations in benchmarks. Introduce a structured generalization score. |
| Robustness in Multi-Turn and Tool-Augmented Settings | Extended dialogues and tool integrations create new jailbreak surfaces, enabling memory poisoning and adversarial drift [19], [49], [50].            | Adopt safety decay metrics and safety memory checkpoints. Incorporate multi-turn tool-use or retrieval-poisoning scenarios into evaluations.       |
| Socio-Technical Challenges and Governance            | Jailbreaks affect trust and compliance; transparency and reproducible testing remain limited [9], [54], [62], [89], [101].                           | Integrate transparent red-teaming and standardized benchmarking into governance frameworks.                                                        |
| Evaluation Standardization and Coverage Gaps         | Benchmarks vary in scope and metrics, limiting comparability across studies [9], [10], [53].                                                         | Define a unified ASR protocol with toxicity-aware and long-context scoring, and expand benchmarks.                                                 |
| Certifiable and Auditable Safety                     | Lack of transparent auditing limits deployment in regulated environments; vulnerabilities persist in domain-specific LLMs [99], [100], [102], [103]. | Develop certifiable alignment frameworks with behavioral versioning and provenance tracking [104], [105].                                          |
| Red-Teaming at Scale                                 | Human red-teaming is costly; automated pipelines adapt slowly to evolving models [5], [8], [54].                                                     | Combine automated attackers with human-in-the-loop refinement for continual red-teaming integrated with defense training [34], [90].               |

rigid reward structures or prompt framing [23]. Adaptive methods, such as controllable safety tuning and goal-prioritization [108], [109], offer flexible moderation tailored to context and domain (e.g., healthcare). Hardening system prompts and integrating preference alignment at inference time could enhance resilience to adversarial reframing.

#### D. Socio-Technical Challenges and Governance

Beyond technical issues, jailbreaks pose socio-technical risks affecting public trust and regulatory compliance [101]. Safety evaluation must include procedural safeguards such as transparent red-teaming, reproducible benchmarks, and risk disclosure [9], [54]. Coordinated governance between researchers, developers, and policymakers can link robustness metrics with accountability standards, improving trust in deployed systems [62], [89].

#### E. Cross-Lingual Safety and Low-Resource Language Bias

Throughout this survey, multiple results indicate that jailbreak and toxicity vulnerabilities are amplified in low-resource languages, dialects, and code-mixed inputs. Multilingual jailbreak and safety studies show that alignment mechanisms trained primarily on English systematically under-detect harmful intent in other languages, especially when prompts use regional varieties, code-switching, or culturally specific metaphors [7], [28], [35], [36], [45]. This pattern reflects not just an additional attack vector but a structural bias in current AI safety research and evaluation pipelines, which concentrate research on English and Western contexts. Future work should treat cross-lingual and cross-cultural robustness as a first-class objective by expanding multilingual red-teaming, curating safety datasets for low-resource languages, and ensuring that benchmark suites such as PromptBench and and, more broadly, that evaluation suites—including PromptBench, JailbreakBench, and multimodal benchmarks such as SneakyPrompt—expand their coverage to encompass linguistic, cultural, and modal diversity [9], [10], [17].

#### F. Evaluation Standardization and Coverage Gaps

Benchmarks like PromptBench and JailbreakBench improve reproducibility but lack comprehensive multimodal and tool-use coverage [10], [17], [110]. Future testbeds should unify text, image, audio, and code evaluations, support automated attacker pipelines, and include longitudinal tracking across model versions. We recommend consolidating benchmark practices under a unified ASR reporting protocol that incorporates toxicity-weighted violation scores and long-context refusal fidelity. Future benchmarks should also integrate multimodal red teaming, automated attacker pipelines, and versioned public leaderboards to support longitudinal robustness tracking across model releases.

#### G. Certifiable and Auditable Safety

LLMs in regulated domains need transparent auditing and behavioral traceability. Recent work highlights vulnerabilities even in fine-tuned clinical models [102]. Emerging methods

such as Concept Enhancement Engineering and Token Highlighter [104], [105] enable interpretable alignment verification. Recent studies on domain-specific LLMs, particularly in safety-critical areas such as medical decision support, have also begun to document domain-specific jailbreak behaviors in vertical models rather than only in general-purpose assistants. Domain-specific evaluations at *NeurIPS 2025*—including *Safe Generative AI*, *CARES*, and *MMed-RAG* [99], [100]—underscore that jailbreak robustness must be assessed in these specialized settings, where misaligned behavior interacts directly with regulatory and safety constraints.

#### H. Red Teaming at Scale

Human red-teaming remains essential but resource-heavy. Automated pipelines like LLMStinger, AutoJailbreak, and AutoDAN [5], [8], [39] improve scalability but lag in adapting to new model architectures. Advanced methods combining gradient-based prompt generation and decoding-level jailbreaks [111], [112] highlight the need for ensemble, adaptive red-teaming. Integrating real-time safety monitoring with iterative training feedback will enable continuous resilience testing.

### X. THREATS TO VALIDITY

While this survey provides a comprehensive synthesis of jailbreak research, several factors may limit completeness and generalizability. The analysis focuses on literature published between 2023 and 2025 across IEEE, arXiv, and ACM repositories, which may exclude earlier or proprietary studies not publicly released. Categorizing attack and defense strategies also involves subjective interpretation, especially in emerging areas such as multi-turn, multimodal, and tool-assisted jailbreaks. Moreover, the jailbreak landscape evolves rapidly; new attack and defense mechanisms appear frequently, which may render some findings temporary. Finally, most studies—and therefore this review—focus on English-language prompts and Western-aligned models, leaving cross-linguistic and cultural robustness less explored. These limitations underscore the need for ongoing, iterative review as methods mature and more diverse datasets become available.

### XI. CONCLUSION AND FUTURE WORK

Jailbreak attacks remain one of the most persistent challenges in ensuring the safe deployment of large language models. This survey consolidates recent developments in attack strategies, defense mechanisms, and evaluation frameworks, situating jailbreaks as a dynamic, optimization-driven phenomenon that evolves alongside generative AI itself. While alignment techniques such as adversarial fine-tuning, RLHF, and benchmark-based evaluation have strengthened resilience, defenses continue to lag behind increasingly adaptive, multimodal, and automated adversarial strategies.

The findings highlight that the arms race between jailbreakers and defenders extends beyond technical solutions—it is also socio-technical. Sustainable safety requires defense frameworks that generalize to unseen adversarial prompts,

operate effectively in multi-turn and tool-augmented environments, and integrate continuous evaluation across diverse models and modalities. Progress also depends on transparent, certifiable, and reproducible safety processes to support accountability and public trust, especially in high-stakes applications such as healthcare, education, and security.

Looking forward, future research should emphasize three key directions: (1) the creation of adaptive safety architectures capable of responding to novel adversarial inputs in real time; (2) development of unified, multimodal benchmarks with consistent, interpretable metrics; and (3) establishment of standardized auditing and provenance frameworks that enable certifiable alignment at scale. By framing jailbreaks as an evolving interaction between optimization and alignment, this survey offers a structured reference for researchers and practitioners seeking to advance robust, transparent, and human-aligned large language models.

TABLE IX  
LIST OF ACRONYMS

| Acronym | Meaning                                    |
|---------|--------------------------------------------|
| LLM     | Large Language Model                       |
| SFT     | Supervised Fine-Tuning                     |
| RLHF    | Reinforcement Learning from Human Feedback |
| ASF     | Adversarial Suffix Filtering               |
| PAIR    | Prompt Automatic Iterative Refinement      |
| RAG     | Retrieval-Augmented Generation             |
| DAN     | Do Anything Now (role-play prompt)         |
| NLP     | Natural Language Processing                |

## REFERENCES

- [1] OpenAI, "Chatgpt," <https://chat.openai.com/>, 2025, accessed: Aug 2025.
- [2] Anthropic PBC, "Claude ai," <https://claude.ai/>, 2025, accessed: Aug 2025.
- [3] Google DeepMind, "Gemini," <https://gemini.google.com/>, 2025, accessed: Aug 2025.
- [4] S. Nabavirazavi, S. Zad, and S. S. Iyengar, "Evaluating the universality of "do anything now" jailbreak prompts on large language models," in *2025 IEEE 15th Annual Computing and Communication Workshop and Conference (CCWC)*, 2025, pp. 00 691–00 696.
- [5] L. Lu, H. Yan, Z. Yuan, J. Shi, W. Wei, P.-Y. Chen, and P. Zhou, "Autojailbreak: Exploring jailbreak attacks and defenses through a dependency lens," 2024. [Online]. Available: <https://arxiv.org/abs/2406.03805>
- [6] D. Khachaturov and R. Mullins, "Adversarial suffix filtering: a defense pipeline for llms," 2025. [Online]. Available: <https://arxiv.org/abs/2505.09602>
- [7] Y. Kumar, C. Paredes, G. Yang, J. J. Li, and P. Morreale, "Adversarial testing of llms across multiple languages," in *2024 International Symposium on Networks, Computers and Communications (ISNCC)*, 2024, pp. 1–6.
- [8] P. Jha, A. Arora, and V. Ganesh, "Llmstinger: Jailbreaking llms using rl fine-tuned llms," 2024. [Online]. Available: <https://arxiv.org/abs/2411.08862>
- [9] K. Zhu, Q. Zhao, H. Chen, J. Wang, and X. Xie, "Promptbench: A unified library for evaluation of large language models," 2024. [Online]. Available: <https://arxiv.org/abs/2312.07910>
- [10] P. Chao, E. DeBenedetti, A. Robey, M. Andriushchenko, F. Croce, V. Schwag, E. Dobriban, N. Flammarion, G. J. Pappas, F. Tramer, H. Hassani, and E. Wong, "Jailbreakbench: An open robustness benchmark for jailbreaking large language models," 2024. [Online]. Available: <https://arxiv.org/abs/2404.01318>
- [11] IEEE, "Ieee xplora digital library," <https://ieeexplore.ieee.org/>, 2025, accessed: Aug 2025.
- [12] Cornell University, "arxiv.org e-print archive," <https://arxiv.org/>, 2025, accessed: Aug 2025.
- [13] Association for Computing Machinery, "Acm digital library," <https://dl.acm.org/>, 2025, accessed: Aug 2025.
- [14] Z. Dong, Z. Zhou, C. Yang, J. Shao, and Y. Qiao, "Attacks, defenses, and evaluations for llm conversation safety: A survey," 2024, nAACL 2024. [Online]. Available: <https://arxiv.org/abs/2402.09283>
- [15] S. Yi, Y. Liu, Z. Sun, T. Cong, X. He, J. Song, K. Xu, and Q. Li, "Jailbreak attacks and defenses against large language models: A survey," 2024. [Online]. Available: <https://arxiv.org/abs/2407.04295>
- [16] A. Zhou, B. Li, and H. Wang, "Robust prompt optimization for defending language models against jailbreaking attacks," 2024. [Online]. Available: <https://arxiv.org/abs/2401.17263>
- [17] Y. Yang, B. Hui, H. Yuan, N. Gong, and Y. Cao, "Sneakyprompt: Jailbreaking text-to-image generative models," 2023. [Online]. Available: <https://arxiv.org/abs/2305.12082>
- [18] Z. Chen, H. Lin, K. Xu, X. Jiang, and T. Sun, "Ghostprompt: Jailbreaking text-to-image generative models based on dynamic optimization," 2025. [Online]. Available: <https://arxiv.org/abs/2505.18979>
- [19] B. Hui, H. Yuan, N. Gong, P. Burlina, and Y. Cao, "Pleak: Prompt leaking attacks against large language model applications," in *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security*, ser. CCS '24. New York, NY, USA: Association for Computing Machinery, 2024, p. 3600–3614. [Online]. Available: <https://doi.org/10.1145/3658644.3670370>
- [20] K. Nakka and N. Saxena, "Bitbypass: A new direction in jailbreaking aligned large language models with bitstream camouflage," arXiv preprint arXiv:2506.02479, 2025, accessed: Aug 2025. [Online]. Available: <https://arxiv.org/abs/2506.02479>
- [21] J. Zhuang, H. Jin, Y. Zhang, Z. Kang, W. Zhang, G. G. Dagher, and H. Wang, "Exploring the vulnerability of the content moderation guardrail in large language models via intent manipulation," 2025. [Online]. Available: <https://arxiv.org/abs/2505.18556>
- [22] D. Pasquini, M. Strohmeier, and C. Troncoso, "Neural exec: Learning (and learning from) execution triggers for prompt injection attacks," 2024. [Online]. Available: <https://arxiv.org/abs/2403.03792>
- [23] J. Su, J. Kempe, and K. Ullrich, "Mission impossible: A statistical perspective on jailbreaking llms," 2024. [Online]. Available: <https://arxiv.org/abs/2408.01420>
- [24] D. Lee, S. Xie, S. Rahman, K. Pat, D. Lee, and Q. A. Chen, "'prompter says': A linguistic approach to understanding and detecting jailbreak attacks against large-language models," in *Proceedings of the 1st ACM Workshop on Large AI Systems and Models with Privacy and Safety Analysis*, ser. LAMPS '24. New York, NY, USA: Association for Computing Machinery, 2024, p. 77–87. [Online]. Available: <https://doi.org/10.1145/3689217.3690618>
- [25] Y. Zeng, H. Lin, J. Zhang, D. Yang, R. Jia, and W. Shi, "How johnny can persuade llms to jailbreak them: Rethinking persuasion to challenge ai safety by humanizing llms," 2024. [Online]. Available: <https://arxiv.org/abs/2401.06373>
- [26] Z. D. Johnson, "Generation, detection, and evaluation of role-play based jailbreak attacks in large language models," Master's thesis, Massachusetts Institute of Technology, 2024, accessed: 29 Aug 2025. [Online]. Available: <https://dspace.mit.edu/handle/1721.1/156989>
- [27] T. T. Wang, J. Hughes, H. Sleight, R. Schaeffer, R. Agrawal, F. Barez, M. Sharma, J. Mu, N. Shavit, and E. Perez, "Jailbreak defense in a narrow domain: Limitations of existing methods and a new transcript-classifier approach," 2024. [Online]. Available: <https://arxiv.org/abs/2412.02159>
- [28] B. Rababah, S. T. Wu, M. Kwiatkowski, C. K. Leung, and C. G. Akcora, "Are existing large language models robust against jailbreak attacks?" in *2024 IEEE International Conference on Big Data (Big-Data)*, 2024, pp. 5383–5391.
- [29] M. Yu, W. Wang, Y. Wei, and S. Qin, "Adaptive jailbreaking strategies based on the semantic understanding capabilities of large language models," 2025. [Online]. Available: <https://arxiv.org/abs/2505.23404>
- [30] V. M. G. Nair and V. V. Dantuluri, "Unmasking the canvas: A dynamic benchmark for image generation jailbreaking and llm content safety," 2025. [Online]. Available: <https://arxiv.org/abs/2505.04146>
- [31] A. Zou, Z. Wang, N. Carlini, M. Nasr, J. Z. Kolter, and M. Fredrikson, "Universal and transferable adversarial attacks on aligned language models," 2023. [Online]. Available: <https://arxiv.org/abs/2307.15043>

- [32] Y. Zeng, Y. Wu, X. Zhang, H. Wang, and Q. Wu, "Autodefense: Multi-agent llm defense against jailbreak attacks," 2024. [Online]. Available: <https://arxiv.org/abs/2403.04783>
- [33] J. Hughes, S. Price, A. Lynch, R. Schaeffer, F. Barez, S. Koyejo, H. Sleight, E. Jones, E. Perez, and M. Sharma, "Best-of-n jailbreaking," 2024. [Online]. Available: <https://arxiv.org/abs/2412.03556>
- [34] S. Jiang, X. Chen, K. Xu, L. Chen, H. Ren, and R. Tang, "Decomposition, synthesis, and attack: A multi-instruction fusion method for jailbreaking llms," *IEEE Internet of Things Journal*, vol. 12, no. 8, pp. 9420–9434, 2025.
- [35] T. S. Luong, T.-T. Le, L. N. Van, and T. H. Nguyen, "Realistic evaluation of toxicity in large language models," 2024. [Online]. Available: <https://arxiv.org/abs/2405.10659>
- [36] R. T. Prome, T. I. Tamiti, and A. Barua, "Leveraging the potential of prompt engineering for hate speech detection in low-resource languages," 2025. [Online]. Available: <https://arxiv.org/abs/2506.23930>
- [37] X. Fu, S. Li, Z. Wang, Y. Liu, R. K. Gupta, T. Berg-Kirkpatrick, and E. Fernandes, "Imprompter: Tricking llm agents into improper tool use," 2024. [Online]. Available: <https://arxiv.org/abs/2410.14923>
- [38] R. Bhardwaj and S. Poria, "Red-teaming large language models using chain of utterances for safety-alignment," 2023. [Online]. Available: <https://arxiv.org/abs/2308.09662>
- [39] S. Zhu, R. Zhang, B. An, G. Wu, J. Barrow, Z. Wang, F. Huang, A. Nenkova, and T. Sun, "Autodan: Interpretable gradient-based adversarial attacks on large language models," 2023. [Online]. Available: <https://arxiv.org/abs/2310.15140>
- [40] K. Takemoto, "All in how you ask for it: Simple black-box method for jailbreak attacks," *Applied Sciences*, vol. 14, no. 9, 2024. [Online]. Available: <https://www.mdpi.com/2076-3417/14/9/3558>
- [41] Y. Huang, Y. Sun, S. Ruan, Y. Zhang, Y. Dong, and X. Wei, "Breaking the ceiling: Exploring the potential of jailbreak attacks through expanding strategy space," 2025. [Online]. Available: <https://arxiv.org/abs/2505.21277>
- [42] M. Russinovich, A. Salem, and R. Eldan, "Great, now write an article about that: The crescendo multi-turn llm jailbreak attack," 2025. [Online]. Available: <https://arxiv.org/abs/2404.01833>
- [43] S. Jiang, X. Chen, and R. Tang, "Deceiving llm through compositional instruction with hidden attacks," *ACM Trans. Auton. Adapt. Syst.*, Feb. 2025, just Accepted. [Online]. Available: <https://doi.org/10.1145/3717067>
- [44] P. Ding, J. Kuang, D. Ma, X. Cao, Y. Xian, J. Chen, and S. Huang, "A wolf in sheep's clothing: Generalized nested jailbreak prompts can fool large language models easily," 2024. [Online]. Available: <https://arxiv.org/abs/2311.08268>
- [45] Z.-X. Yong, C. Menghini, and S. H. Bach, "Low-resource languages jailbreak gpt-4," 2024. [Online]. Available: <https://arxiv.org/abs/2310.02446>
- [46] X. Qi, K. Huang, A. Panda, P. Henderson, M. Wang, and P. Mittal, "Visual adversarial examples jailbreak aligned large language models," 2023. [Online]. Available: <https://arxiv.org/abs/2306.13213>
- [47] C. W. Chiu, L. Huang, B. Li, H. Chen, and K.-K. R. Choo, "'do as i say not as i do': A semi-automated approach for jailbreak prompt attack against multimodal llms," 2025. [Online]. Available: <https://arxiv.org/abs/2502.00735>
- [48] D. Yao, J. Zhang, I. G. Harris, and M. Carlsson, "Fuzzllm: A novel and universal fuzzing framework for proactively discovering jailbreak vulnerabilities in large language models," in *ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, Apr. 2024, p. 4485–4489. [Online]. Available: <http://dx.doi.org/10.1109/ICASSP48485.2024.10448041>
- [49] S. Kim, Y. Lee, Y. Song, and K. Lee, "What really matters in many-shot attacks? an empirical study of long-context vulnerabilities in llms," 2025. [Online]. Available: <https://arxiv.org/abs/2505.19773>
- [50] H. Hu, A. Robey, and C. Liu, "Steering dialogue dynamics for robustness against multi-turn jailbreaking attacks," 2025. [Online]. Available: <https://arxiv.org/abs/2503.00187>
- [51] S. Irtiza, K. A. Akbar, A. Yasmeen, L. Khan, O. Daescu, and B. Thuraisingham, "LLM-sentry: A model-agnostic human-in-the-loop framework for securing large language models," in *2024 IEEE 6th International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA)*, 2024, pp. 245–254.
- [52] H. Jin, R. Chen, P. Zhang, A. Zhou, Y. Zhang, and H. Wang, "Guard: Role-playing to generate natural-language jailbreakings to test guideline adherence of large language models," 2025. [Online]. Available: <https://arxiv.org/abs/2402.03299>
- [53] H. Cai, A. Arunasalam, L. Y. Lin, A. Bianchi, and Z. B. Celik, "Rethinking how to evaluate language model jailbreak," 2024. [Online]. Available: <https://arxiv.org/abs/2404.06407>
- [54] Z. Wang, C. Q. Knight, J. Kritz, W. E. Primack, and J. Michael, "A red teaming roadmap towards system-level safety," arXiv preprint arXiv:2506.05376, 2025, accessed: 29 Aug 2025. [Online]. Available: <https://arxiv.org/abs/2506.05376>
- [55] W. Lin, A. Gerchanovsky, O. Akgul, L. Bauer, M. Fredrikson, and Z. Wang, "Llm whisperer: An inconspicuous attack to bias llm responses," 2025. [Online]. Available: <https://arxiv.org/abs/2406.04755>
- [56] X. Zhang, C. Zhang, T. Li, Y. Huang, X. Jia, M. Hu, J. Zhang, Y. Liu, S. Ma, and C. Shen, "Jailguard: A universal detection framework for llm prompt-based attacks," 2025. [Online]. Available: <https://arxiv.org/abs/2312.10766>
- [57] S. Casper, J. Lin, J. Kwon, G. Culp, and D. Hadfield-Menell, "Explore, establish, exploit: Red teaming language models from scratch," 2023. [Online]. Available: <https://arxiv.org/abs/2306.09442>
- [58] Y. Tao, Y. Shen, H. Zhang, Y. Shen, L. Wang, C. Shi, and S. Du, "Robustness of large language models against adversarial attacks," 2024. [Online]. Available: <https://arxiv.org/abs/2412.17011>
- [59] X. Chen, Y. Nie, L. Yan, Y. Mao, W. Guo, and X. Zhang, "RL-jack: Reinforcement learning-powered black-box jailbreaking attack against llms," 2024. [Online]. Available: <https://arxiv.org/abs/2406.08725>
- [60] V. Benjamin, E. Braca, I. Carter, H. Kanchwala, N. Khojasteh, C. Landow, Y. Luo, C. Ma, A. Magarelli, R. Mirin, A. Moyer, K. Simpson, A. Skawinski, and T. Heverin, "Systematically analyzing prompt injection vulnerabilities in diverse llm architectures," 2024.
- [61] B. Upadhyay and V. Behzadan, "Sandwich attack: Multi-language mixture adaptive attack on llms," in *Proceedings of the 4th Workshop on Trustworthy Natural Language Processing (TrustNLP 2024)*. Association for Computational Linguistics, 2024, p. 208–226. [Online]. Available: <http://dx.doi.org/10.18653/v1/2024.trustnlp-1.18>
- [62] B. Wang, W. Chen, H. Pei, C. Xie, M. Kang, C. Zhang, C. Xu, Z. Xiong, R. Dutta, R. Schaeffer, S. T. Truong, S. Arora, M. Mazeika, D. Hendrycks, Z. Lin, Y. Cheng, S. Koyejo, D. Song, and B. Li, "Decodingtrust: A comprehensive assessment of trustworthiness in gpt models," 2024. [Online]. Available: <https://arxiv.org/abs/2306.11698>
- [63] J. Shi, Z. Yuan, Y. Liu, Y. Huang, P. Zhou, L. Sun, and N. Z. Gong, "Optimization-based prompt injection attack to llm-as-a-judge," 2025. [Online]. Available: <https://arxiv.org/abs/2403.17710>
- [64] A. Yadav, H. Jin, M. Luo, J. Zhuang, and H. Wang, "Inflood: Jailbreaking large language models with information overload," 2025. [Online]. Available: <https://arxiv.org/abs/2506.12274>
- [65] H. Liang, Y. Sun, Y. Cai, J. Zhu, and B. Zhang, "Jailbreaking llms' safeguard with universal magic words for text embedding models," 2025. [Online]. Available: <https://arxiv.org/abs/2501.18280>
- [66] E. Bethany, M. Bethany, J. A. N. Flores, S. K. Jha, and P. Najafirad, "Jailbreaking large language models with symbolic mathematics," 2024. [Online]. Available: <https://arxiv.org/abs/2409.11445>
- [67] M. Kim, H. Lee, B. Gong, H. Zhang, and S. J. Hwang, "Automatic jailbreaking of the text-to-image generative ai systems," 2024. [Online]. Available: <https://arxiv.org/abs/2405.16567>
- [68] A. Mehrotra, M. Zampetakis, P. Kassinik, B. Nelson, H. Anderson, Y. Singer, and A. Karbasi, "Tree of attacks: Jailbreaking black-box llms automatically," 2024. [Online]. Available: <https://arxiv.org/abs/2312.02119>
- [69] Y. Xue, J. Wang, Z. Yin, Y. Ma, H. Qin, R. Tao, and X. Liu, "Dual intention escape: Penetrating and toxic jailbreak attack against large language models," in *Proceedings of the ACM on Web Conference 2025*, ser. WWW '25. New York, NY, USA: Association for Computing Machinery, 2025, p. 863–871. [Online]. Available: <https://doi.org/10.1145/3696410.3714654>
- [70] M. Arazzi, V. K. Kembu, A. Nocera, and V. P. "Xbreaking: Explainable artificial intelligence for jailbreaking llms," 2025. [Online]. Available: <https://arxiv.org/abs/2504.21700>
- [71] P. Chao, A. Robey, E. Dobriban, H. Hassani, G. J. Pappas, and E. Wong, "Jailbreaking black box large language models in twenty queries," 2024. [Online]. Available: <https://arxiv.org/abs/2310.08419>
- [72] G. Deng, Y. Liu, Y. Li, K. Wang, Y. Zhang, Z. Li, H. Wang, T. Zhang, and Y. Liu, "Masterkey: Automated jailbreaking of large language model chatbots," in *Proceedings 2024 Network and Distributed System Security Symposium*. Internet Society, 2024. [Online]. Available: <http://dx.doi.org/10.14722/ndss.2024.24188>
- [73] A. Wei, N. Haghtalab, and J. Steinhardt, "Jailbroken: How does llm safety training fail?" 2023. [Online]. Available: <https://arxiv.org/abs/2307.02483>

- [74] M. Sharma, M. Tong, J. Mu, J. Wei, J. Kruthoff, S. Goodfriend, E. Ong, A. Peng, R. Agarwal, C. Anil, A. Askell, N. Bailey, J. Benton, E. Bluemke, S. R. Bowman, E. Christiansen, H. Cunningham, A. Dau, A. Gopal, R. Gilson, L. Graham, L. Howard, N. Kalra, T. Lee, K. Lin, P. Lofgren, F. Mosconi, C. O'Hara, C. Olsson, L. Petrini, S. Rajani, N. Saxena, A. Silverstein, T. Singh, T. Summers, L. Tang, K. K. Troy, C. Weissner, R. Zhong, G. Zhou, J. Leike, J. Kaplan, and E. Perez, "Constitutional classifiers: Defending against universal jailbreaks across thousands of hours of red teaming," 2025. [Online]. Available: <https://arxiv.org/abs/2501.18837>
- [75] Z. Xu, F. Jiang, L. Niu, J. Jia, B. Y. Lin, and R. Poovendran, "Safedecoding: Defending against jailbreak attacks via safety-aware decoding," 2024. [Online]. Available: <https://arxiv.org/abs/2402.08983>
- [76] E. Galinkin and M. Sablotny, "Improved large language model jailbreak detection via pretrained embeddings," 2024. [Online]. Available: <https://arxiv.org/abs/2412.01547>
- [77] Z. Jin, S. Liu, H. Li, X. Zhao, and H. Qu, "Jailbreakhunter: A visual analytics approach for jailbreak prompts discovery from large-scale human-llm conversational datasets," 2024. [Online]. Available: <https://arxiv.org/abs/2407.03045>
- [78] Z. Wei, Y. Liu, and N. B. Erichson, "Emoji attack: Enhancing jailbreak attacks against judge llm detection," 2025. [Online]. Available: <https://arxiv.org/abs/2411.01077>
- [79] Z. Sun, S. Shen, S. Cao, H. Liu, C. Li, Y. Shen, C. Gan, L.-Y. Gui, Y.-X. Wang, Y. Yang, K. Keutzer, and T. Darrell, "Aligning large multimodal models with factually augmented rlhf," 2023. [Online]. Available: <https://arxiv.org/abs/2309.14525>
- [80] Z. Zhou, H. Yu, X. Zhang, R. Xu, F. Huang, and Y. Li, "How alignment and jailbreak work: Explain llm safety through intermediate hidden states," 2024. [Online]. Available: <https://arxiv.org/abs/2406.05644>
- [81] B. Upadhyay, V. Behzadan, and A. Karbasi, "Cognitive overload attack: prompt injection for long context," 2024. [Online]. Available: <https://arxiv.org/abs/2410.11272>
- [82] A. Kumar, C. Agarwal, S. Srinivas, A. J. Li, S. Feizi, and H. Lakkaraju, "Certifying llm safety against adversarial prompting," 2025. [Online]. Available: <https://arxiv.org/abs/2309.02705>
- [83] H. Kim, S. Yuk, and H. Cho, "Break the breakout: Reinventing lm defense against jailbreak attacks with self-refinement," 2024. [Online]. Available: <https://arxiv.org/abs/2402.15180>
- [84] P. Kumar, E. Lau, S. Vijayakumar, T. Trinh, S. R. Team, E. Chang, V. Robinson, S. Hendryx, S. Zhou, M. Fredrikson, S. Yue, and Z. Wang, "Refusal-trained llms are easily jailbroken as browser agents," 2024. [Online]. Available: <https://arxiv.org/abs/2410.13886>
- [85] M. Andriushchenko, A. Souly, M. Dziemian, D. Duenas, M. Lin, J. Wang, D. Hendrycks, A. Zou, Z. Kolter, M. Fredrikson, E. Winsor, J. Wynne, Y. Gal, and X. Davies, "Agentharm: A benchmark for measuring harmfulness of llm agents," 2025. [Online]. Available: <https://arxiv.org/abs/2410.09024>
- [86] Y. Wu, X. Li, Y. Liu, P. Zhou, and L. Sun, "Jailbreaking gpt-4v via self-adversarial attacks with system prompts," 2024. [Online]. Available: <https://arxiv.org/abs/2311.09127>
- [87] S. Chen, A. Zharmagambetov, S. Mahloujifar, K. Chaudhuri, D. Wagner, and C. Guo, "Secalign: Defending against prompt injection with preference optimization," arXiv preprint arXiv:2410.05451, 2025. [Online]. Available: <https://arxiv.org/abs/2410.05451>
- [88] G. Cornacchia, G. Zizzo, M. Fraser, M. Z. Hameed, A. Rawat, and M. Purcell, "Moje: Mixture of jailbreak experts, naive tabular classifiers as guard for prompt attacks," 2024. [Online]. Available: <https://arxiv.org/abs/2409.17699>
- [89] B. Rababah, Shang, Wu, M. Kwiatkowski, C. Leung, and C. G. Akcora, "Sok: Prompt hacking of large language models," 2024. [Online]. Available: <https://arxiv.org/abs/2410.13901>
- [90] Y. Liu, G. Deng, Z. Xu, Y. Li, Y. Zheng, Y. Zhang, L. Zhao, T. Zhang, K. Wang, and Y. Liu, "Jailbreaking chatgpt via prompt engineering: An empirical study," 2024. [Online]. Available: <https://arxiv.org/abs/2305.13860>
- [91] R. Li, H. Wang, and C. Mao, "Largo: Latent adversarial reflection through gradient optimization for jailbreaking llms," 2025. [Online]. Available: <https://arxiv.org/abs/2505.10838>
- [92] Y. Wang, Z. Shi, A. Bai, and C.-J. Hsieh, "Defending llms against jailbreaking attacks via backtranslation," 2024. [Online]. Available: <https://arxiv.org/abs/2402.16459>
- [93] W. Mai, G. Hong, P. Chen, X. Pan, B. Liu, Y. Zhang, H. Duan, and M. Yang, "You can't eat your cake and have it too: The performance degradation of llms with jailbreak defense," 2025. [Online]. Available: <https://arxiv.org/abs/2501.12210>
- [94] H. Strohmer, Y. Dasri, and D. Murzello, "Exploring security vulnerabilities in chatgpt through multi-technique evaluation of resilience to jailbreak prompts and defensive measures," in *2024 International Conference on Computer and Applications (ICCA)*, 2024, pp. 1–12.
- [95] C. Pathade, "Red teaming the mind of the machine: A systematic evaluation of prompt injection and jailbreak vulnerabilities in llms," 2025. [Online]. Available: <https://arxiv.org/abs/2505.04806>
- [96] W. Luo, S. Ma, X. Liu, X. Guo, and C. Xiao, "Jailbreakv: A benchmark for assessing the robustness of multimodal large language models against jailbreak attacks," 2024. [Online]. Available: <https://arxiv.org/abs/2404.03027>
- [97] J. Jeong, S. Bae, Y. Jung, J. Hwang, and E. Yang, "Playing the fool: Jailbreaking llms and multimodal llms with out-of-distribution strategy," 2025. [Online]. Available: <https://arxiv.org/abs/2503.20823>
- [98] J. Ma, Y. Li, Z. Xiao, A. Cao, J. Zhang, C. Ye, and J. Zhao, "Jailbreaking prompt attack: A controllable adversarial attack against diffusion models," 2025. [Online]. Available: <https://arxiv.org/abs/2404.02928>
- [99] S. Chen, X. Li, M. Zhang, E. H. Jiang, Q. Zeng, and C.-H. Yu, "Cares: Comprehensive evaluation of safety and adversarial robustness in medical llms," 2025. [Online]. Available: <https://arxiv.org/abs/2505.11413>
- [100] P. Xia, K. Zhu, H. Li, T. Wang, W. Shi, S. Wang, L. Zhang, J. Zou, and H. Yao, "Mmed-rag: Versatile multimodal rag system for medical vision language models," 2025. [Online]. Available: <https://arxiv.org/abs/2410.13085>
- [101] C. Barrett, B. Boyd, E. Bursztein, N. Carlini, B. Chen, J. Choi, A. R. Chowdhury, M. Christodorescu, A. Datta, S. Feizi, K. Fisher, T. Hashimoto, D. Hendrycks, S. Jha, D. Kang, F. Kerschbaum, E. Mitchell, J. Mitchell, Z. Ramzan, K. Shams, D. Song, A. Taly, and D. Yang, "Identifying and mitigating the security risks of generative ai," *Foundations and Trends® in Privacy and Security*, vol. 6, no. 1, p. 1–52, 2023. [Online]. Available: <http://dx.doi.org/10.1561/33000000041>
- [102] H. Zhang, Q. Lou, and Y. Wang, "Towards safe ai clinicians: A comprehensive study on large language model jailbreaking in healthcare," 2025. [Online]. Available: <https://arxiv.org/abs/2501.18632>
- [103] S. Cohen, R. Bitton, and B. Nassi, "A jailbroken genai model can cause substantial harm: Genai-powered applications are vulnerable to promptwares," 2024. [Online]. Available: <https://arxiv.org/abs/2408.05061>
- [104] J. Yang, Z. Lin, Z. Lu, Y. Wang, L. Wang, T. Wei, X. Du, and S. Yang, "Cee: An inference-time jailbreak defense for embodied intelligence via subspace concept rotation," 2025. [Online]. Available: <https://arxiv.org/abs/2504.13201>
- [105] X. Hu, P.-Y. Chen, and T.-Y. Ho, "Token highlighter: Inspecting and mitigating jailbreak prompts for large language models," 2024. [Online]. Available: <https://arxiv.org/abs/2412.18171>
- [106] A. Robey, Z. Ravichandran, V. Kumar, H. Hassani, and G. J. Pappas, "Jailbreaking llm-controlled robots," 2024. [Online]. Available: <https://arxiv.org/abs/2410.13691>
- [107] A. M. Shibli, M. M. A. Pritom, and M. Gupta, "Abusegpt: Abuse of generative ai chatbots to create smishing campaigns," 2024. [Online]. Available: <https://arxiv.org/abs/2402.09728>
- [108] J. Zhang, A. Elgohary, A. Magooda, D. Khashabi, and B. V. Durme, "Controllable safety alignment: Inference-time adaptation to diverse safety requirements," 2025. [Online]. Available: <https://arxiv.org/abs/2410.08968>
- [109] Z. Zhang, J. Yang, P. Ke, F. Mi, H. Wang, and M. Huang, "Defending large language models against jailbreaking attacks through goal prioritization," 2024. [Online]. Available: <https://arxiv.org/abs/2311.09096>
- [110] T. Zhang, B. Cao, Y. Cao, L. Lin, P. Mitra, and J. Chen, "Wordgame: Efficient i& effective llm jailbreak via simultaneous obfuscation in query and response," 2024. [Online]. Available: <https://arxiv.org/abs/2405.14023>
- [111] Y. Huang, S. Gupta, M. Xia, K. Li, and D. Chen, "Catastrophic jailbreak of open-source llms via exploiting generation," 2023. [Online]. Available: <https://arxiv.org/abs/2310.06987>
- [112] Z. Yu, X. Liu, S. Liang, Z. Cameron, C. Xiao, and N. Zhang, "Don't listen to me: Understanding and exploring jailbreak prompts of large language models," 2024. [Online]. Available: <https://arxiv.org/abs/2403.17336>